13D research: ML, NLP and long-term active risk

Investment aim: use machine learning (ML) and natural language processing (NLP) to uncover long-term alpha signals in public companies backed by high-quality event-driven or activist hedge funds, then take active risk in several selected, potentially concentrated positions.

Status: the historical simple-following baseline has not established a reliable investable edge. Filing preparation and manager scorecards are underway; predictive ML and incremental intent NLP have not been evaluated.

The intended output is a ranked set of company opportunities with an economic thesis, catalyst, downside case and position rationale. Manager reputation and a disclosed plan are candidate inputs; their value must be tested using information available before purchase.

1. Three phases - baseline, selection and disclosed intentions

Phase / purpose	Information and comparison	What has been done
1. Simple following Measure post-disclosure returns.	Initial 13D events and manager groups; stock returns versus Russell size-segment ETF proxies.	Historical baseline evaluated. No reliable edge established; some funded simulations incomplete.
2. Structured ML Rank stronger opportunities.	Investor history, ownership, company size, liquidity, volatility and prior returns; fundamentals planned.	15,813 candidates audited; HFR scorecards prepared. Dated joins/features unfinished; no fitted model.
3. Incremental NLP Add the investor's stated plan.	Item 4 objectives and specificity; compare structured ML with ML plus text on the same sample.	Planned. No validated intent/alpha result. Ownership/identity extraction is preparation work.

2. Data sources and their role

Source	Information	Use / status
SEC EDGAR	13D/13D-A documents, public acceptance times, filer/issuer identities, stakes and Item 4 text.	Cached source-audited filing archive; factual extraction and future intent NLP. [E2]
CRSP via WRDS	Daily stock/ETF prices, returns, volume, historical security IDs and delisting fields.	Market data behind the reported historical return tests and simulations. [E1, E4]
FTSE Russell via WRDS	Historical Russell constituent snapshots; ETF returns come from CRSP.	Stock-appropriate size-segment ETF benchmark assignments. [E4]
HFR	35 supplied quarterly snapshots, 2018 Q1-2026 Q3: fund metadata, strategy, reported returns and assets.	Manager candidates and descriptive peer scorecards; predictive value untested. [E3]
Compustat via WRDS	Fundamentals for valuation, leverage, profitability and balance-sheet features.	Planned features. A historically available, joined ML panel is unfinished. [E4]

WRDS = Wharton Research Data Services, the access platform. **Yahoo Finance** supplied an early ten-company price-only preview, separate from these CRSP-based results; the preview script used Yahoo directly rather than yfinance.

HFR fund performance is different from target-stock returns available to an outside follower. Dated manager identities, availability and incomplete dead-fund coverage need validation before the scorecards can become signals. [E3]

Proposed extension: amendments for position adjustments

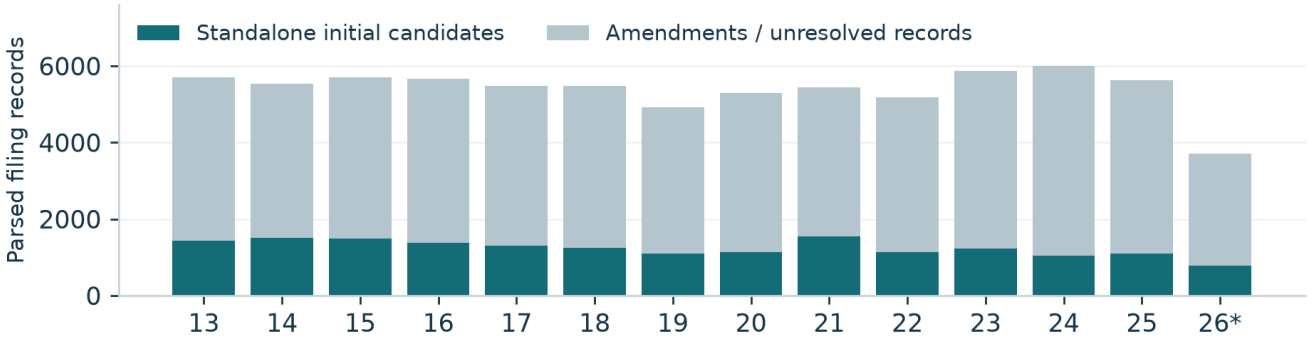
Amendment disclosure	Potential use and observability
Stake increases / reductions	Compare Item 5 shares, issuer ownership percentages and transactions. Percentage changes may reflect issuer share-count changes; ownership percentage is not fund portfolio weight.
Exit / reporting threshold	Item 5(e) dates a fall to 5% or less, which is not necessarily zero. Confirm a full disposition explicitly or leave the full-exit date unknown. [E5]

From about 75,000 filings to 15,813 candidates

Cached SEC archive: January 2, 2013-September 11, 2026 | Unit: parsed filing records

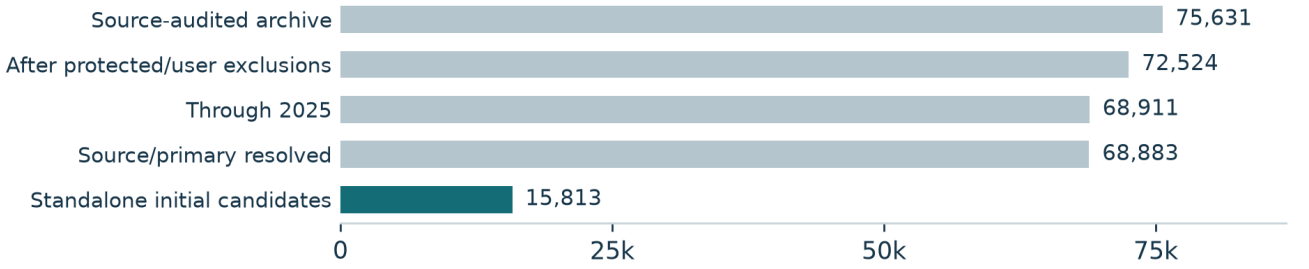
The audited archive contains **75,631 records**, including **58,166 amendments**, **17,420 standalone initial-filing candidates** and **45 unresolved records**. These are filing records, not unique companies, managers or independently validated campaigns. [E2]

3. Archive coverage by acceptance year



Years shown as 13-26. *2026 is partial through September 11. The initial-candidate segment is before the staging exclusions below. This is the project's cached, parsed archive, not a completeness claim about every SEC filing. [E2]

4. The preparation filter - counts reconcile in priority order



Step / reason	Removed	Remaining
Source-audited archive	-	75,631
Protect review records/issuers; apply other user exclusions	3,107	72,524
Restrict acceptance dates to December 31, 2025 or earlier	3,613	68,911
Remove unresolved source/primary-document cases	28	68,883
Retain standalone initial candidates; exclude remaining other records	53,070	**15,813**

Each record is counted once under the first applicable exclusion. The final 53,070 includes remaining amendments and other non-initial records; it is not the full archive's amendment count. The staging run preserves review protections and prior user exclusions. [E2]

What the 15,813 represent: initial-filing candidates accepted in 2013-2025, all processed by the factual staging audit. They are not yet a validated hedge-fund universe or a tradable ML panel. Protected review samples remain independent.

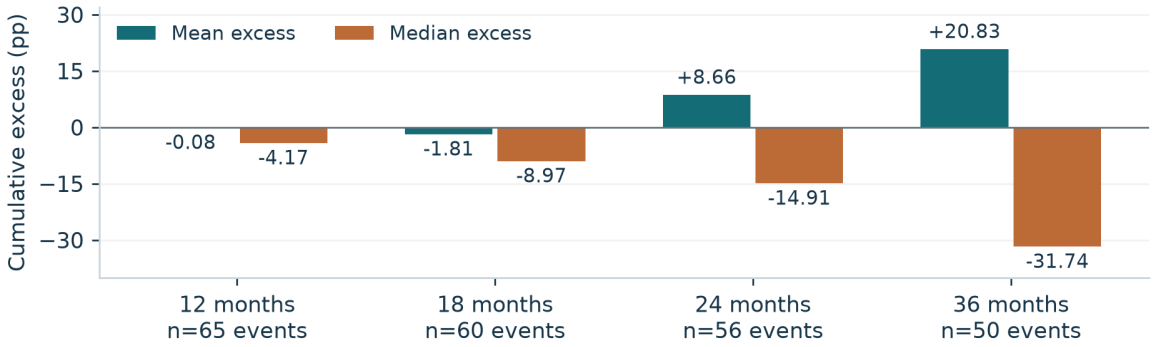
Amendments remain useful: the 58,166 amendment records are retained in the archive. They are excluded from the initial-entry cohort, but can support a separately tested strategy for adding, trimming or exiting after public updates.

Separate return-study sample: Phase 1 uses a 2018-2025 schedule and additional security, benchmark and outcome checks. At 24 months, the broad sample is 2,107 scheduled > 712 benchmark-assigned > 694 calculable; its focused activist-candidate subset is 67 > 58 > 56. The 56 events are not a direct funnel from the later 15,813-record staging exercise. [E1]

Historical returns: positive averages, weak medians

Phase 1 exploratory development | 2018-2025 research window | Focused activist-candidate subset

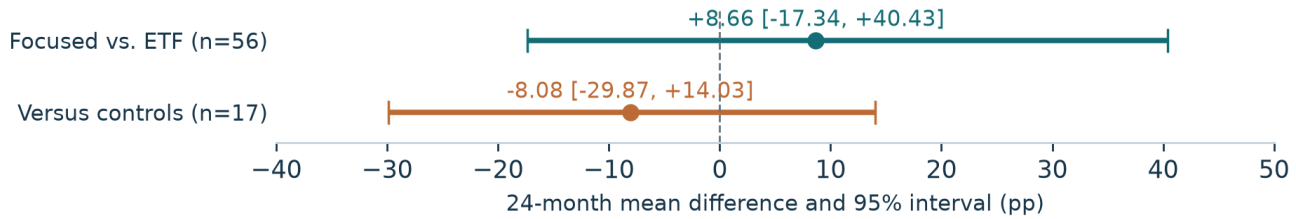
5. The mean improves with horizon; the median remains negative



Hold	n	Stock mean	ETF mean	Excess mean	Excess median	Beat ETF	Mean excess 95% interval
12 mo	65	+11.58%	+11.65%	-0.08 pp	-4.17 pp	47.7% (31/65)	-11.25 to +13.64 pp
18 mo	60	+13.52%	+15.34%	-1.81 pp	-8.97 pp	35.0% (21/60)	-19.82 to +23.45 pp
24 mo	56	+27.82%	+19.15%	+8.66 pp	-14.91 pp	41.1% (23/56)	-17.34 to +40.43 pp
36 mo	50	+53.57%	+32.74%	+20.83 pp	-31.74 pp	32.0% (16/50)	-20.87 to +84.21 pp

Gross, cumulative, equally weighted event returns; excess = stock minus its Russell size-segment ETF proxy; pp = percentage points. Horizons use different, overlapping calculable samples, so higher means do not establish an optimal holding period. Intervals use the saved issuer bootstrap. [E1]

6. Two-year results weaken under a matched comparison



Both 95% intervals cross zero. In 17 industry/year/size-matched cases, focused-event excess averaged -3.07 pp versus +5.01 pp for controls: difference -8.08 pp. This subset differs from the 56-event ETF comparison; the two estimates must not be pooled. [E1]

7. A funded account is a separate test

Completed $100k / 12-month account	Result	Comparison / context
March 19, 2018-December 31, 2025	+57.41% cumulative +5.99% annualized	Cash-only +22.87%; SPY +181.73%; IWV +166.93% cumulative.
Deployment and downside	75 buys; 55.41% average cash -35.15% maximum drawdown	5% issuer targets, max. 20; cash interest, liquidity limits and modeled costs. Different exposures prevent alpha claims.

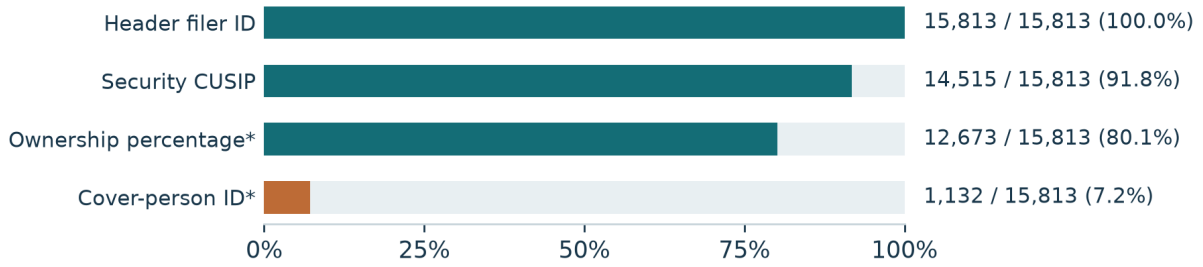
Conventions: modeled entry at the opening price on the second exchange session after SEC acceptance; 252/378/504/756 sessions approximate 12/18/24/36 months. Observed opens do not guarantee fills. No simultaneous duplicate-issuer positions in funded simulations. Historical stock/benchmark coverage is selective; monthly membership dates have availability limits.

Primary $100k 18/24/36-month accounts remain incomplete at Aratana's replacement-share/contingent-right settlement; proceeds are not invented. The gross event table does not resolve that account issue. Holding horizons were revised after inspecting earlier results; this is exploratory evidence. [E1]

From prepared data to an investable selection test

Next research steps | Candidate mechanisms, not demonstrated alpha signals

8. Extraction coverage is substantial; economic links need work



Denominator: 15,813 selected 2013-2025 filings. *At least one extracted value per filing; this does not establish that every person or ownership row was recovered. Bars overlap. Header filer ID is not a validated fund identity; a CUSIP is not a verified historical stock link. Coverage measures presence, not accuracy. [E2]

9. Proposed signals and their investment mechanism

Candidate signal	Economic hypothesis	Evidence needed
Manager quality / history Dated HFR peers; prior target-stock outcomes.	Repeated execution skill may distinguish managers whose plans create value.	Validate historical identities and availability; separate fund returns from outside-follower returns.
Company economics Valuation, leverage, cash flow, liquidity.	A credible catalyst may have more upside where value and financial capacity support it.	Build dated fundamentals/market features; test against size, industry and risk explanations.
Disclosed intent (NLP) Board change, asset sale, capital return; plan specificity.	A concrete, feasible plan may change cash flows or capital allocation beyond what prices reflect.	Validate text labels against source spans; compare ML with and without NLP on identical events.
Amendment-driven adjustments Stake increases, trims, changed plans and disclosed exits.	New disclosures may change conviction, position size or the case for continuing to hold.	Link consecutive filings; separate full exits from 5%-or-less disclosures. Trade after public acceptance; test against fixed holding periods.

10. Validation gates before taking concentrated active risk

Next step	What it must establish
Make information historically available	Resolve manager/stock links and missingness; use only facts public before purchase. Freeze targets, manager-quality rules and sample definitions.
Test incremental selection value	Use chronological evaluation and common samples/trading costs. Compare simple following, structured ML and ML plus NLP; report uncertainty, downside and coverage.
Translate conviction into several positions	Evaluate liquidity, loss scenarios, shared sector/factor/manager exposures and position sizes. Concentrated sizing is proposed; existing 5% targets are a baseline.

Amendment chains and full-exit dates are not yet validated. Keep transaction dates separate from SEC disclosure dates; simulated adjustments must follow public acceptance. Previously inspected history is development data; freeze a protocol and reserve an unexposed or prospective evaluation before claiming a repeatable edge.

Evidence key: E1: Sept 24 long-horizon summaries, matched controls and Phase 1 conclusion. E2: universe ledger/staging selection and coverage audit. E3: Sept 30 HFR protocol/scorecards. E4: CRSP/FTSE source inventory, WRDS records and Yahoo preview. E5: SEC guidance (103.08, 104.05, 110.05/08) and 2023 final rule. Local source links in companion notes.

Prepared October 4, 2026 from preserved project evidence. AI-assisted synthesis and graphics; no new backtest, trained model, intent labels or changes to independent human-review materials.